

These papers explain

- What to do after detection
- Risk assessment
- Alert generation
- Smart guidance

★ If this were my project, I would actually choose these five research areas:

Section	Search Term	Why
Paper 1	Sound Event Detection	Build the AI model
Paper 2	TinyML / Edge AI for Audio	Run efficiently on phones
Paper 3	Acoustic Scene Classification / Audio Scene Understanding	Differentiate dangerous vs non-dangerous situations
Paper 4	DCASE / AudioSet / FSD50K	Select and evaluate datasets
Paper 5	Emergency Decision Support Systems	Generate intelligent actions after detection

This combination will make your literature survey much stronger than using five papers that all solve the same sound detection problem. It also mirrors the architecture of your proposed system: **detect** → **understand context** → **run efficiently on-device** → **rely on quality datasets** → **provide emergency guidance**.

   ...  Sources

Today 8:19 PM

give complete project first to end explanation divide it into 10 weeks plan and give abstract of project

Project Title

Echo: AI-Based Hazardous Sound Detection and Smart Emergency Guidance System

1. Problem Statement

Every day, incidents like robbery, gun violence, assault, fire accidents, domestic violence, road accidents, and natural disasters occur without immediate assistance. Although smartphones are equipped with microphones, GPS, AI chips, and internet connectivity, they are rarely used as **proactive safety devices**.

Existing emergency applications require users to manually press an SOS button. In many real emergencies, victims may not have the opportunity to do so.

Echo aims to transform smartphones into an **AI-powered personal safety assistant** that continuously monitors environmental sounds, intelligently identifies hazardous situations, verifies them using contextual information, and provides appropriate emergency guidance.

2. Objective

Develop an AI-powered mobile application capable of:

- Detecting hazardous sounds in real time.
- Running efficiently on smartphones.
- Distinguishing between dangerous and non-dangerous situations.
- Alerting users only when confidence is sufficiently high.
- Providing intelligent emergency guidance.
- (Future scope) Notifying nearby users or emergency services.

3. Motivation

Imagine someone:

- Walking home at night
- Alone in a parking lot
- Traveling
- Elderly
- Child

The phone hears

- Gunshots
- Screams
- Explosion
- Glass breaking

Instead of simply notifying

The application should say

Possible gunfire detected nearby.

Recommended action:

- Stay behind concrete cover.
- Avoid open roads.
- Nearest Police Station: 600m.
- Emergency Number: 112

4. Proposed System



5. Modules

Module 1

Audio Capture

Continuously records short audio windows.

Example

5-second audio clip

↓

Process

↓

Delete

No permanent recording.

Only temporary buffers.

Module 2

Audio Preprocessing

Converts audio into AI-friendly features.

Examples

- Mel Spectrogram
- MFCC
- Log Mel Spectrogram

Noise reduction

Voice enhancement

Normalization

Module 3

Hazardous Sound Detection

This is the core AI.

Detects

- Gunshot
- Explosion
- Glass breaking

- Fire alarm
- Police siren
- Ambulance siren
- Distress scream
- Dog attack

Model

CNN

or

CNN + Transformer

Output

Gunshot

Confidence

97%

Module 4

Context Verification

The most important module.

Instead of

Gunshot detected

↓

Alert

The AI asks

Was it really dangerous?

Checks

Bluetooth speaker?

Movie?

TV?

Inside theatre?

Multiple dangerous sounds?

Continuous screaming?

Running detected?

Time?

Location?

Confidence

Gunshot

96%

Movie detected

↓

Final confidence

21%

↓

Ignore

Module 5

Emergency Decision Engine

After danger confirmation

Generate actions.

Examples

Gunshot

↓

Hide



Nearest Police Station

Explosion



Move Away



Hospital

Fire



Nearest Exit



Fire Station

Robbery



Share Live Location



Call 112

Module 6

Smart Navigation

Uses

GPS

Google Maps

Suggests

Nearest

- Police Station
- Hospital
- Mall
- Metro
- Public Area

Module 7

Notification

Notify

User

↓

Emergency Contact

↓

(Optional)

Nearby Users

6. Technologies

Frontend

Flutter

Backend

FastAPI

AI

TensorFlow

PyTorch

TensorFlow Lite

Database

Firebase

Maps

Google Maps API

Notifications

Firebase Cloud Messaging

7. AI Models

Initially

CNN

Later

CRNN

Transformer

Future

BEATs

AST

CLAP

8. Datasets

AudioSet

UrbanSound8K

ESC-50

FSD50K

DCASE

DASE

9. Novelty

Most existing systems

Sound



Alert

Echo

Sound



Context



Verify



Danger Score



Emergency Guidance

This significantly reduces false alarms.

10. Future Scope

Wearables

Smartwatch

IoT

Smart Cities

Police Integration

Edge AI

Drone Integration

11. Expected Output

Example

Gunshot Detected

Confidence

96%

Nearby Police Station

350m

Hospital

1.2km

Suggested Action

Stay Low

Hide Behind Concrete

Avoid Open Roads

Call 112

12. 10-Week Development Plan

Week 1

Literature Survey

Study

- Sound Event Detection
- TinyML
- Edge AI
- Audio Scene Classification
- Emergency Systems

Deliverables

- Literature review
- Problem statement
- Research gap

Week 2

Dataset Collection

Download

- AudioSet
- UrbanSound8K
- ESC-50
- FSD50K

Select

Dangerous sounds

Safe sounds

Prepare dataset.

Week 3

Data Preprocessing

Convert audio

↓

Mel Spectrogram

↓

MFCC

↓

Training data

Perform augmentation

- Noise addition
- Pitch shift
- Time stretch

Week 4

Train AI Model

Train

CNN

Evaluate

Accuracy

Precision

Recall

F1 Score

Confusion Matrix

Week 5

Context Verification Module

Design logic

Example

Movie?

Bluetooth Speaker?

Running?

Location?

Reduce false positives.

Week 6

Emergency Decision Engine

Design rule engine

Sound

↓

Recommended Action



Safety Tips



Emergency Numbers

Week 7

Mobile Application

Develop Flutter app

Features

Microphone

Notification

GPS

Google Maps

UI

Week 8

AI Integration

TensorFlow Lite

Run AI locally

Real-time detection

Optimize inference speed.

Week 9

Testing

Test

Gunshots

Sirens

Movies

TV

Traffic

Measure

Latency

Battery

False positives

False negatives

Week 10

Finalization

Prepare

Presentation

Report

Demo

Poster

Video

Project documentation

Future Research (Final Year Extension)

For a major project or publication, you could extend Echo by adding:

- **Multimodal sensing:** combine audio with accelerometer, gyroscope, GPS, Bluetooth context, and camera (with user permission).

- **Speech understanding:** recognize emergency phrases like "Help!", "Call the police!", or explicit threats.
- **Crowd verification:** require multiple nearby devices to independently detect the same event before sending community alerts.
- **Federated learning:** improve the model across users without uploading raw audio, preserving privacy.
- **Offline Edge AI:** optimize models with TensorFlow Lite or ONNX Runtime for continuous on-device inference.
- **Smartwatch integration:** trigger alerts from both phone and wearable.
- **Official emergency integration:** notify emergency contacts or authorities where APIs and regulations permit.

Abstract

Echo: AI-Based Hazardous Sound Detection and Smart Emergency Guidance System is an intelligent mobile safety application that leverages edge artificial intelligence to identify hazardous acoustic events in real time and assist users during emergencies. The system continuously analyzes ambient audio using lightweight deep learning models running directly on the device to detect events such as gunshots, distress screams, glass breaking, explosions, and emergency sirens. To minimize false alarms, Echo incorporates context-aware reasoning by considering acoustic patterns, device state, and environmental cues before classifying an event as hazardous. Once a threat is confirmed, the application provides situation-specific guidance, identifies nearby safe locations using GPS and mapping services, and can notify trusted contacts. By combining real-time sound event detection, context verification, edge AI, and intelligent decision support, Echo aims to transform smartphones into proactive personal safety assistants while maintaining user privacy through on-device processing. The proposed system is designed to achieve low latency, efficient resource utilization, and reliable emergency assistance, making it suitable as both a research-oriented academic project and a foundation for future smart public safety applications.

